

Commerce XI Mathematics & Statistics - Part 2

Unit 2: Measures of Dispersion

Formulae, Methods and Interpretation

A quick reference based on textbook pages 24-36.

How to use this guide

Step	What to check
1. Identify	Raw data, frequency data, grouped data, combined groups, or comparison by C.V.
2. Calculate	Choose range, Q.D., variance, S.D., combined S.D. or C.V. as requested.
3. Interpret	Smaller dispersion or C.V. means greater consistency; larger C.V. means greater variability.

Scope

Range - quartile deviation - variance - standard deviation - change of origin and scale - combined groups - coefficient of variation

1. Basic measures

Measure	Formula	Use
Range	$L - S$	Quick spread from largest and smallest values
Interquartile range	$Q3 - Q1$	Spread of the central 50%
Quartile deviation	$(Q3 - Q1)/2$	Useful for open-end distributions
Coefficient of Q.D.	$(Q3 - Q1)/(Q3 + Q1)$	Relative quartile spread

2. Variance and standard deviation

Data	Mean	Variance
Raw	$\bar{x} = \sum(x)/n$	$\text{Var}(X) = \sum(x^2)/n - (\bar{x})^2$
Frequency	$\bar{x} = \sum(fx)/N$	$\text{Var}(X) = \sum(fx^2)/N - (\bar{x})^2$
Standard deviation	$S.D. = \sigma$	$\sigma = \text{positive square root of } \text{Var}(X)$

Always square the mean after calculating it. Standard deviation has the original unit; variance has the squared unit.

3. Assumed mean and step deviation

Method	Definition	Result
Assumed mean	$d = X - A$	$\text{Var}(X) = \frac{\sum(f d^2)}{N} - \left\{ \frac{\sum(f d)}{N} \right\}^2$
Step deviation	$u = (X - A)/h$	$\text{Var}(X) = h^2 \left[\frac{\sum(f u^2)}{N} - \left\{ \frac{\sum(f u)}{N} \right\}^2 \right]$
Change of scale	$X = A + hu$	$\text{S.D.}(X) = h \text{S.D.}(u)$

4. Combined groups

Quantity	Formula
Combined size	$N = n_1 + n_2$
Combined mean	$\bar{x} = \frac{(n_1 \bar{x}_1 + n_2 \bar{x}_2)}{(n_1 + n_2)}$
Combined variance	$\frac{[n_1\{\sigma_1^2 + d_1^2\} + n_2\{\sigma_2^2 + d_2^2\}]}{(n_1 + n_2)}$
Offsets	$d_1 = \bar{x}_1 - \text{combined mean}; d_2 = \bar{x}_2 - \text{combined mean}$

5. Coefficient of variation

Formula	Interpretation
$\text{C.V.} = (\text{S.D.}/\text{mean}) \times 100$	A relative, unit-free percentage measure
Smaller C.V.	More consistent or more stable
Larger C.V.	More variable

6. Verified textbook-answer notes

Exercise 2.3, Q1

The combined mean is 44. The combined variance is 55.6, so the combined S.D. is $\sqrt{55.6}$, approximately 7.46. The printed/extracted key text '55.6516' combines 55.6 and 7.46; it is not $\sqrt{55.6516}$.

Miscellaneous Exercise 2, Q15

$CV(\text{Brand-I}) = 22.22\%$ and $CV(\text{Brand-II}) = 20.83\%$. Therefore Brand-I is more variable and Brand-II is more consistent. Any key statement calling Brand-II more variable conflicts with the calculated values and the C.V. rule.

7. Final checks

Check	Action
Square root	Take the positive square root only when converting variance to S.D.
Percent answer	Include the % sign when the question asks for C.V. or a percentage.
Comparison	Compare C.V., rather than S.D. alone, when means or units differ.
Rounding	Round only at the requested final stage.
Combined data	Include both within-group variance and between-mean differences.